



CrysDiS

Crystal Diffraction Simulator

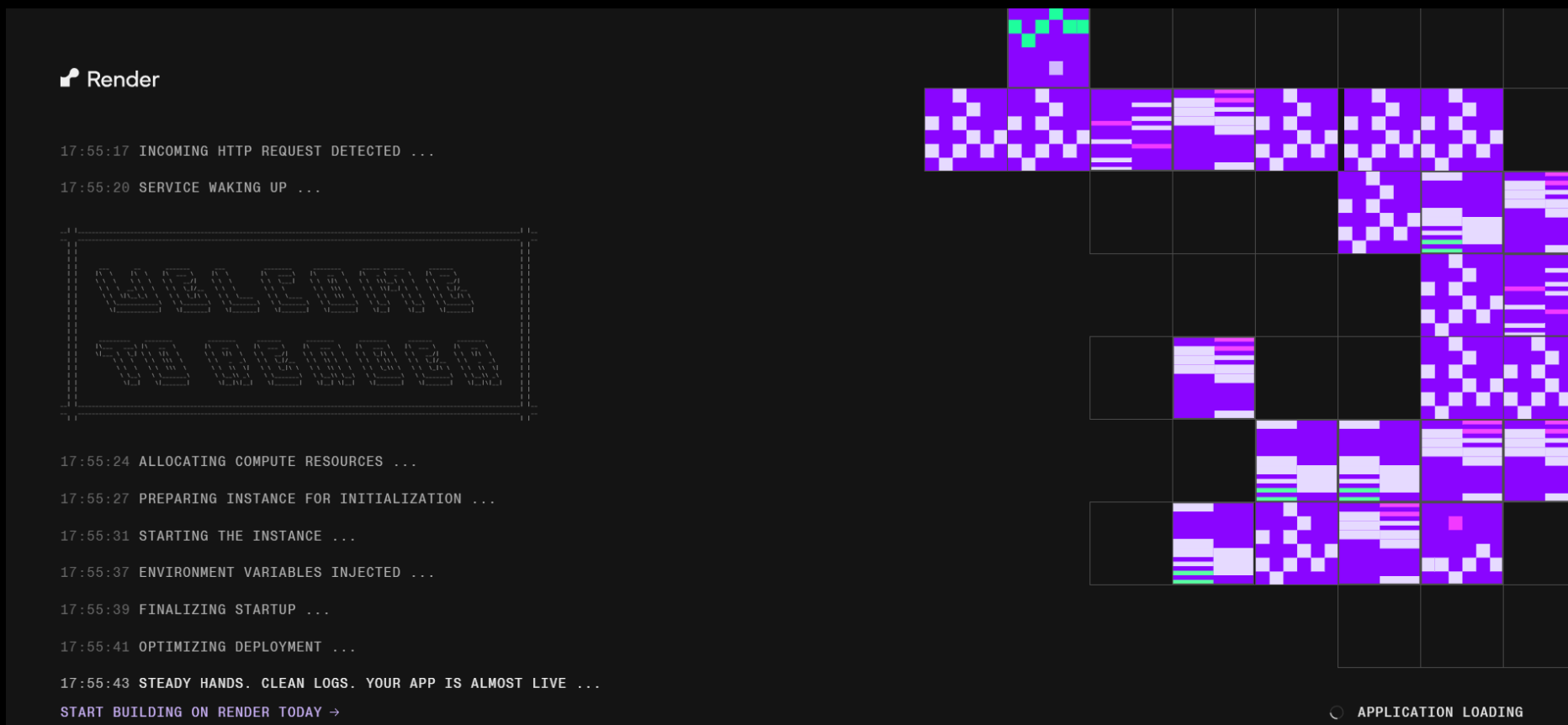
(created using ChatGPT Codex)



Start the program

Option 1: use the website version

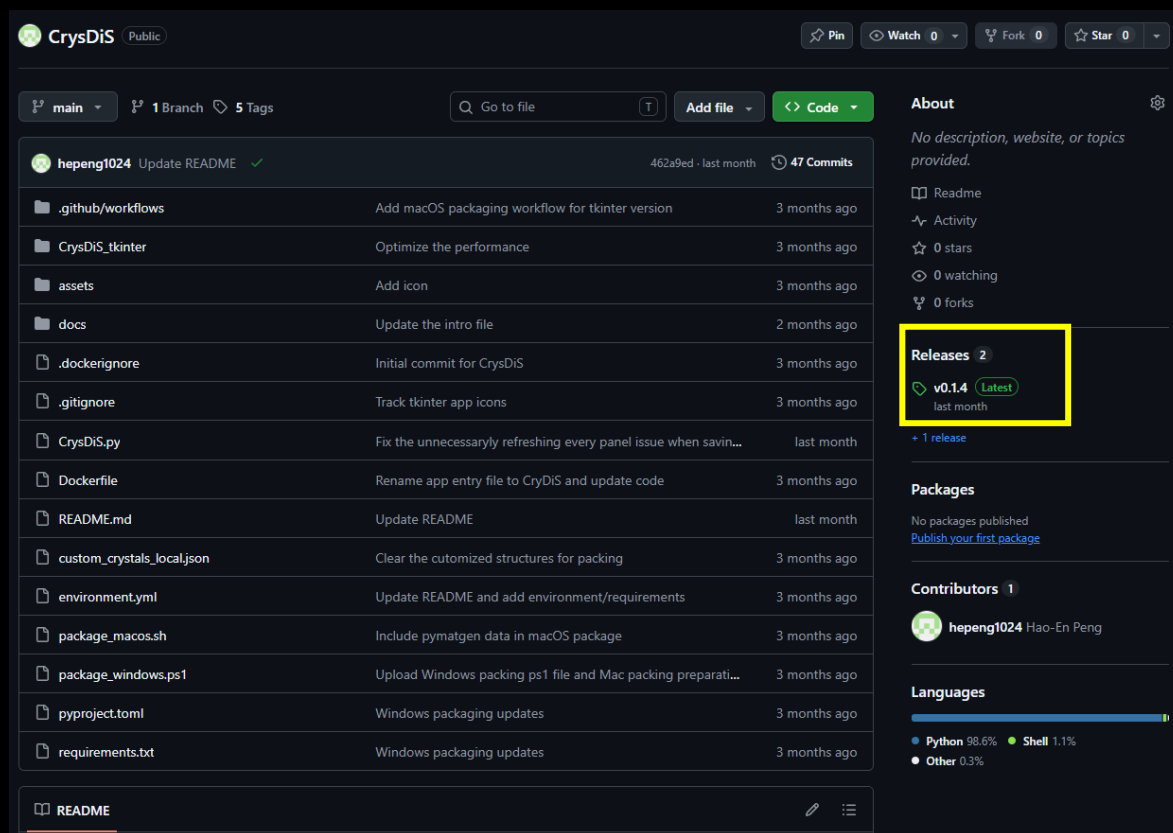
- Website: <https://crysdys.onrender.com/>
- Since this is a free URL, it may take 2~3 minutes to warm up
- It also does not have high computational performance
- If you think this software is useful, running it locally is strongly recommended



Start the program

Option 2: download the program and run it locally

- Go to <https://github.com/hepeng1024/CrysDiS> → Click “Releases” → Download the version that matches your OS



CrysDiS Public

main 1 Branch 5 Tags

Go to file Add file Code

Releases 2

v0.1.4 (Latest) last month

+ 1 release

Packages

No packages published

[Publish your first package](#)

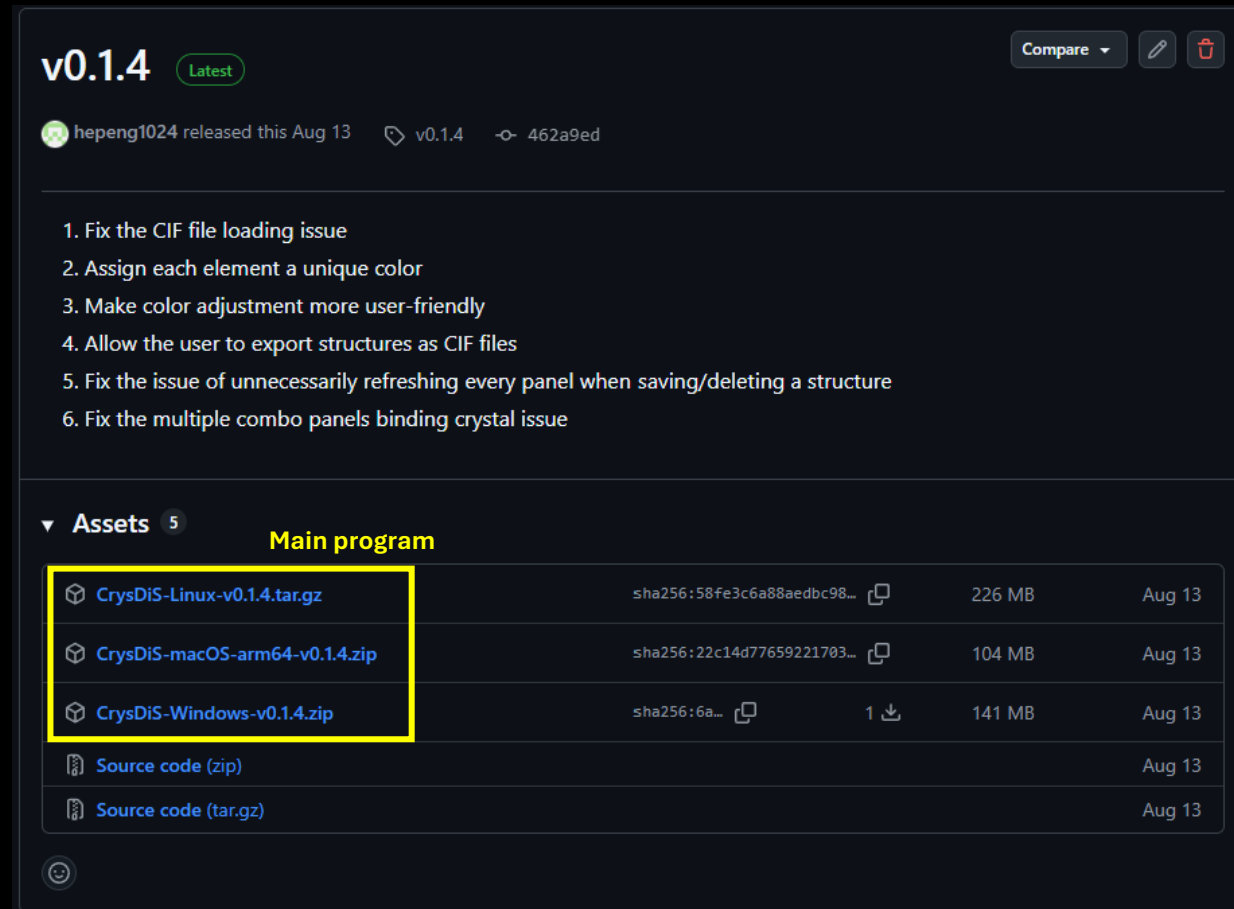
Contributors 1

hepeng1024 Hao-En Peng

Languages

Python 98.6% Shell 1.1% Other 0.3%

README



v0.1.4 (Latest)

hepeng1024 released this Aug 13 v0.1.4 462a9ed

Assets 5

Main program

Asset	SHA256	Size	Date
CrysDiS-Linux-v0.1.4.tar.gz	sha256:58fe3c6a88aedbc98...	226 MB	Aug 13
CrysDiS-macOS-arm64-v0.1.4.zip	sha256:22c14d77659221703...	104 MB	Aug 13
CrysDiS-Windows-v0.1.4.zip	sha256:6a...	141 MB	Aug 13
Source code (zip)			Aug 13
Source code (tar.gz)			Aug 13

- Please refer to the README file if you need more information

Start the program

Option 3: Run the source code from GitHub

- First install Git and Anaconda/Miniconda. Then clone the repository:

```
git clone https://github.com/hepeng1024/CrysDiS.git
```

```
cd CrysDiS
```

- Create the conda environment:

```
conda env create -f environment.yml
```

```
conda activate crysdis
```

- Start CrysDiS:

```
python CrysDiS.py
```

- Then open this address in a browser:

```
http://127.0.0.1:8080
```

- If port 8080 is already being used, choose another port:

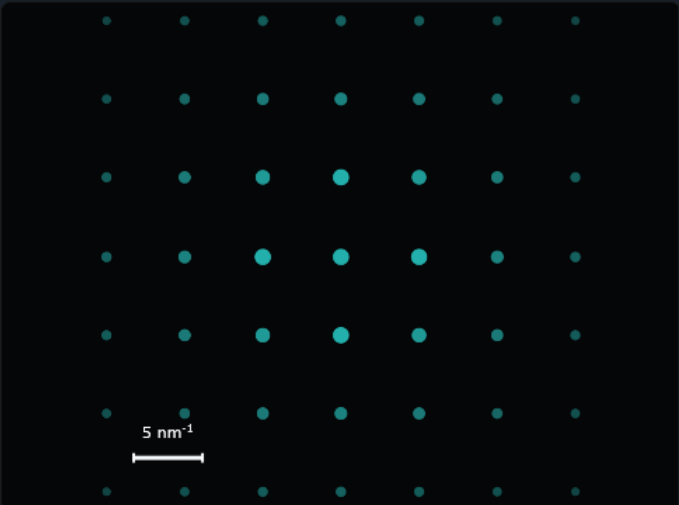
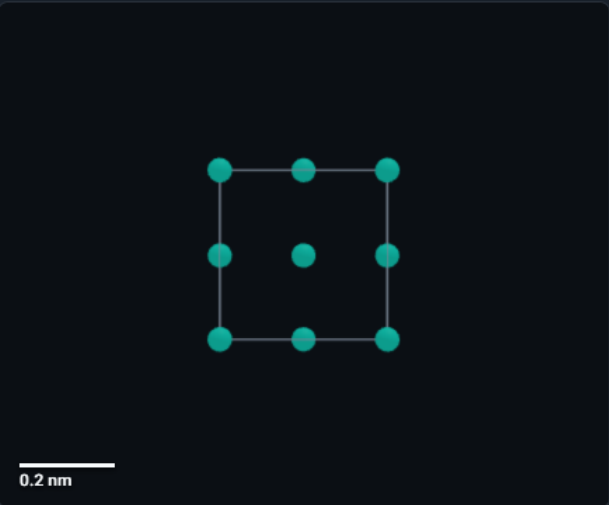
```
NICEGUI_PORT=8094 python CrysDiS.py
```

- Please refer to the README file at <https://github.com/hepeng1024/CrysDiS> for more details

Starting interface

CrysDiS [? INTRO](#) ☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar [ADVANCED](#) [CRYSTAL LIST](#) [LOAD CIF](#) [NEW CRYSTAL](#) [ADD COMBO PANEL](#) [ADD PANEL](#)

1 Crystal: FCC Zone axis: 100 Plane: Vector: Rotation: [APPLY](#) [SYNC](#)   



Features of the top toolbar

“ADD PANEL” allows you to have multiple structures at the same time

CrysDIS ? INTRO

☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar **ADVANCED** **CRYSTAL LIST** **LOAD CIF** **NEW CRYSTAL** **ADD COMBO PANEL** **ADD PANEL**

1 Crystal: FCC Zone axis: 100 Plane: Vector: Rotation: **APPLY** **SYNC** [Camera] [Pencil] [Hand] [Close]

0.2 nm

5 nm⁻¹

2 Crystal: BCC Zone axis: 100 Plane: Vector: Rotation: **APPLY** **SYNC** [Camera] [Pencil] [Hand] [Close]

0.2 nm

5 nm⁻¹

3 Crystal: HCP Zone axis: 100 Plane: Vector: Rotation: **APPLY** **SYNC** [Camera] [Pencil] [Hand] [Close]

0.5 nm

5 nm⁻¹

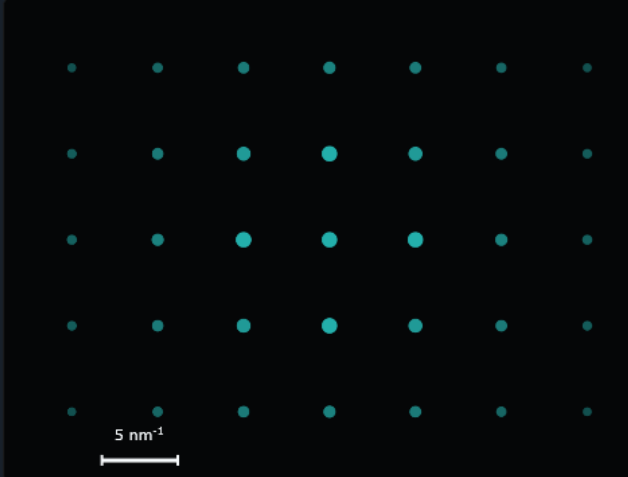
Features of the top toolbar

“ADD COMBO PANEL” allows you to overlap diffraction patterns from different panels and compare them

CrysDis ? INTRO

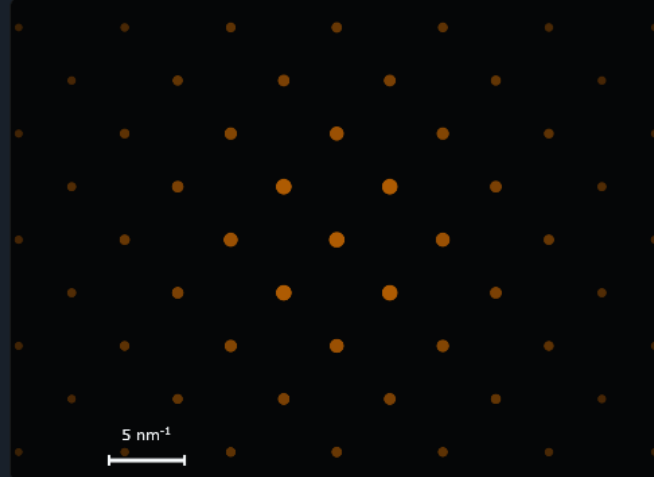
☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar **ADVANCED** **CRYSTAL LIST** **LOAD CIF** **NEW CRYSTAL** **ADD COMBO PANEL** **ADD PANEL**

1 Crystal: FCC Zone axis: 100 Plane: Vector: Rotation: **APPLY** **SYNC**



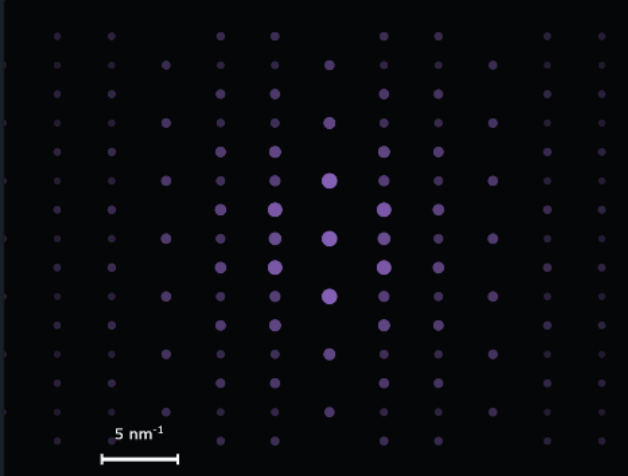
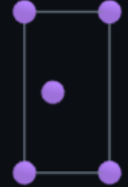
0.2 nm

2 Crystal: BCC Zone axis: 100 Plane: Vector: Rotation: **APPLY** **SYNC**



0.2 nm

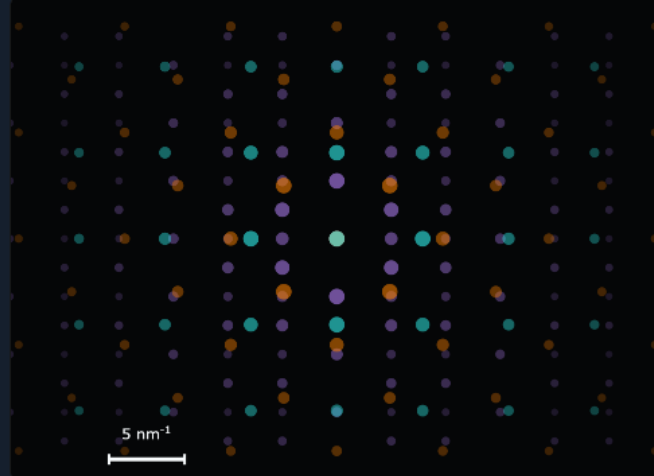
3 Crystal: HCP Zone axis: 100 Plane: Vector: Rotation: **APPLY** **SYNC**



0.5 nm

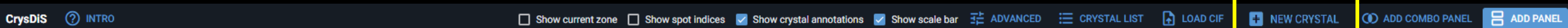
C1 Current crystals: 1. FCC **ADD** **REMOVE** ☐ Blind crystal motion

Panel	Crystal	Zone	
1	FCC	[100]	
2	BCC	[100]	
3	HCP	[2-1-10]	



Features of the top toolbar

“NEW CRYSTAL” lets you build your own crystal



Customized crystal builder



Name Customized crystal	Lattice cubic	Symmetry 1: P1	Save target User app-data library	a 0.35 nm
b 0.35 nm	c 0.35 nm	alpha 90 deg	beta 90 deg	gamma 90 deg

 APPLY LATTICE CONSTRAINTS	 ADD ATOM SITE	 PREVIEW SYMMETRY	 APPLY SAME COLOR TO SAME ELEMENT	 SET ELEMENT COLOR TO DEFAULT
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→ Apply the symmetry to the current structure

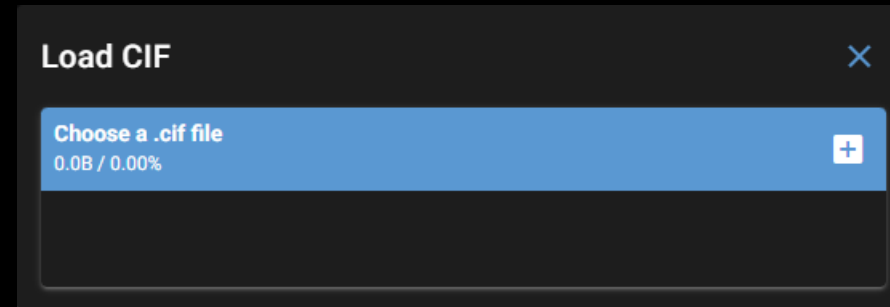
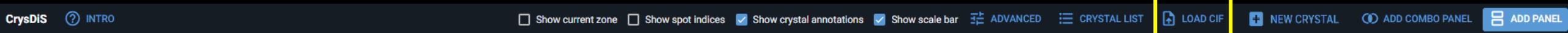
Atom sites

Atom Ni	x 0	y 0	z 0	Occ. 1	Color #ed02e9	Site Ni
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 SAVE AS A NEW STRUCTURE	 EXPORT AS CIF
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Features of the top toolbar

You can load your own .cif files via “LOAD CIF”



Features of the top toolbar

Manage the crystal structures via “CRYSTAL LIST”

CrysDiS

INTRO

Show current zone

Show spot indices

Show crystal annotations

Show scale bar

ADVANCED

CRYSTAL LIST

LOAD CIF

NEW CRYSTAL

ADD COMBO PANEL

ADD PANEL

Three built-in structures that cannot be edited

Crystal list			
FCC	Built-in · cubic, Fm-3m, 1 site(s)	  	
BCC	Built-in · cubic, Im-3m, 1 site(s)	  	
HCP	Built-in · hexagonal, P1, 2 site(s)	  	
D024 (Ni,Mn,Fe)3Ti	Custom · hexagonal, P6_3/mmc, 8 site(s)	  	
D024 (Ni,Mn,Fe)3Ti_unequal_site_occ	Custom · hexagonal, P6_3/mmc, 8 site(s)	  	
D024 (Ni0.8Mn0.1Fe0.1)3Ti	Custom · hexagonal, P6_3/mmc, 8 site(s)	  	
D024 Ni3(Fe,Mn,Ti)	Custom · hexagonal, P6_3/mmc, 8 site(s)	  	
D024 Ni3(Fe,Mn,Ti)_unequal_site_occ	Custom · hexagonal, P6_3/mmc, 8 site(s)	  	
D024 Ni3(Fe0.1Mn0.1Ti0.8)	Custom · hexagonal, P6_3/mmc, 8 site(s)	  	
D024 Ni3Ti	Custom · hexagonal, P6_3/mmc, 4 site(s)	  	
D024 Ni3Ti (disordered)	Custom · hexagonal, P6_3/mmc, 8 site(s)	  	
D024 Ni3Ti (matproj_unsymmetrized)	Custom · hexagonal, P1, 16 site(s)	  	
D024 Ni3Ti (matproj_unsymmetrized distorted)		  	

Features of the top toolbar

Advanced settings in “ADVANCED”

CrysDiS [?](#) [INTRO](#) ☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar **ADVANCED** [CRYSTAL LIST](#) [LOAD CIF](#) [NEW CRYSTAL](#) [ADD COMBO PANEL](#) [ADD PANEL](#)

Advanced settings

Simulation method
Pymatgen TEMCalculator

Camera length
200 mm

Voltage
500 kV

Thickness
10 nm

Max hkl
7

Spot intensity threshold
0.001

Compression factor
100
Contrast of diffraction spots intensity

Spot size scaling
0.3
How large bright spots are

☒ Always snap back view
Check to rotate the crystal back when moving to a new zone

☒ Use four-index basis for hexagonal systems

☐ Auto sync crystal/diffraction

[PALETTE FOR VECTORS/PLANES](#)

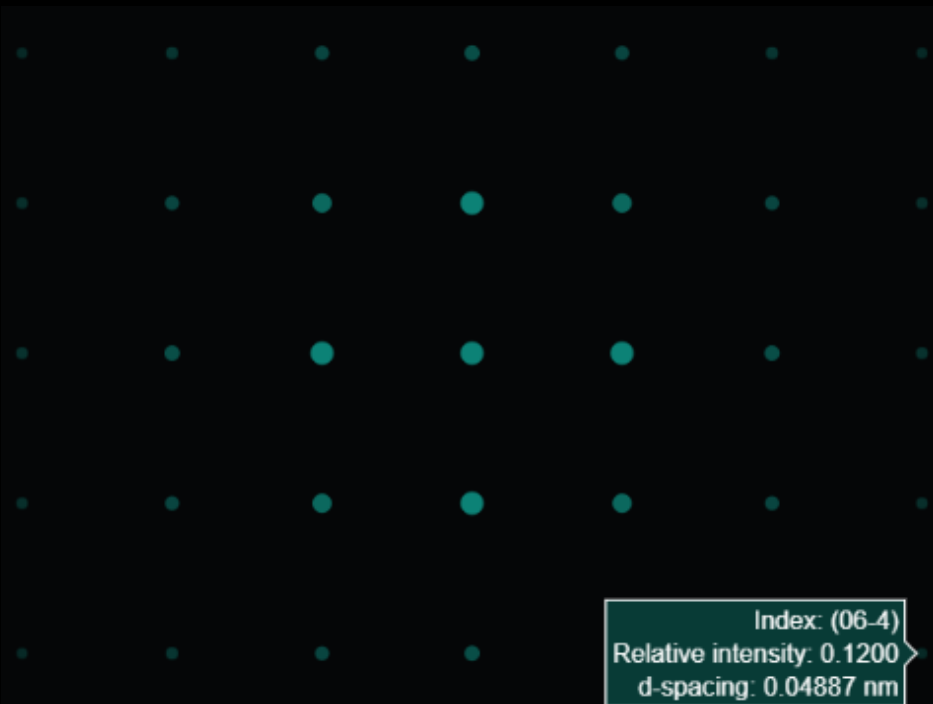
[DIFFRACTION PATTERN COLORS](#)

Log
Ready
Added a comparison panel
Added a comparison panel
Added a combo panel

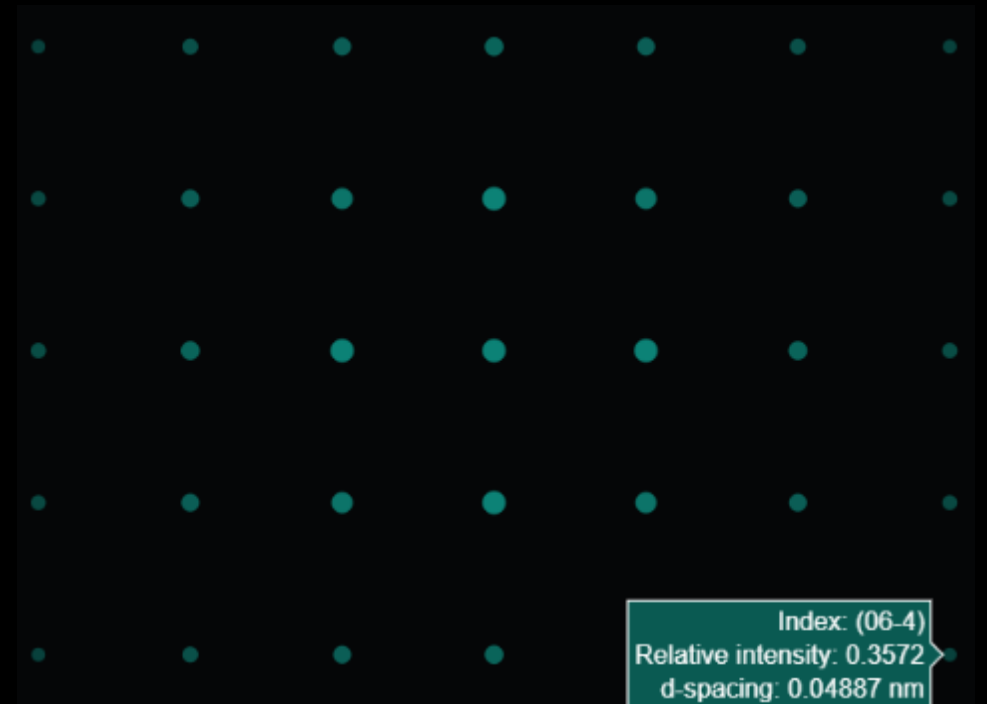
More about advanced settings

- Compression factor

Compression factor = 10



Compression factor = 150



Dim spots are more visible

More about advanced settings

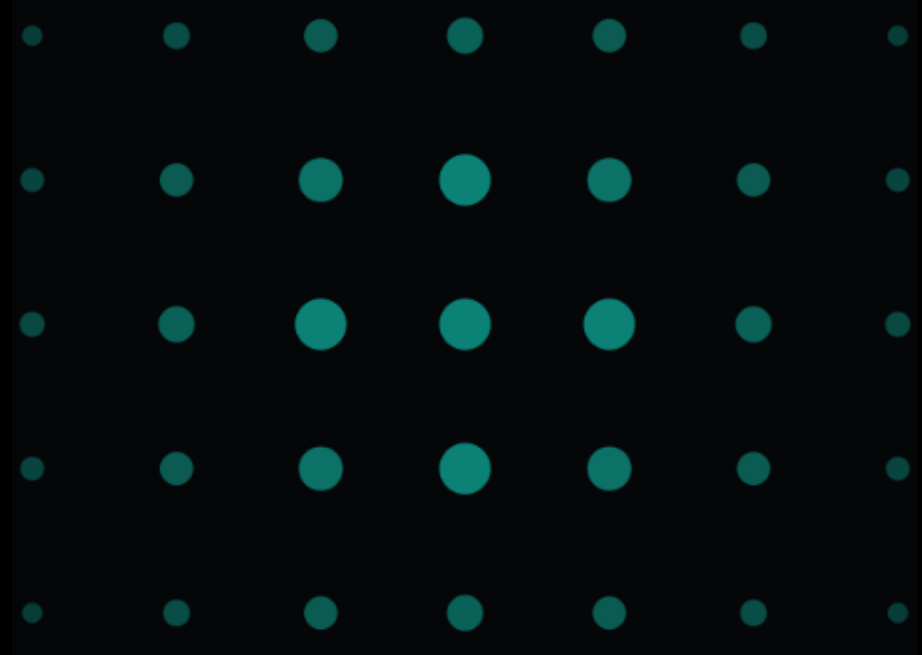
- Spot size scaling effect

Scaling factor = 0



All spots are the same size

Scaling factor = 1



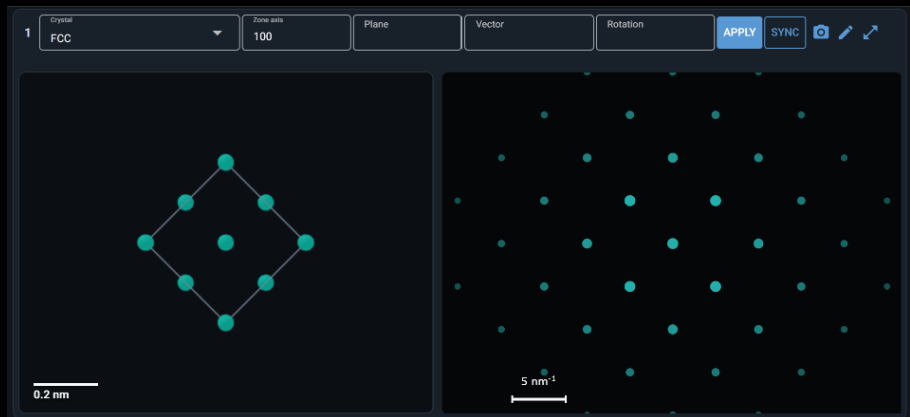
Spot size scales significantly with intensity

More about advanced settings

- Always snap back view



In-plane rotation by 45 deg

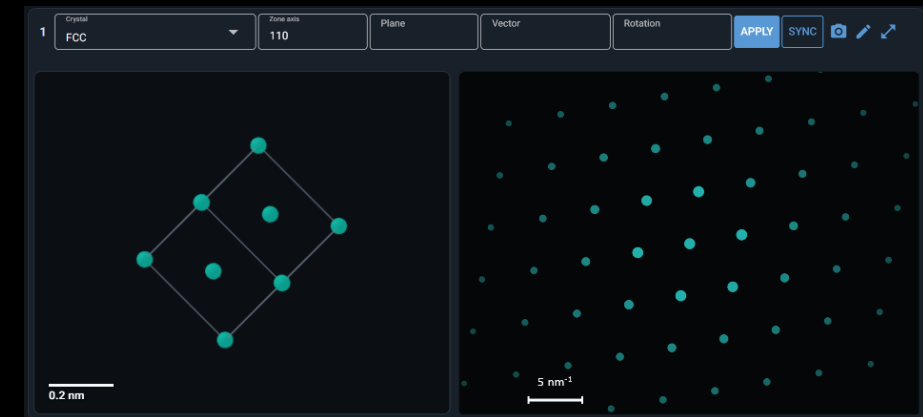


Go to the 110 zone

“Always snap back view” is checked



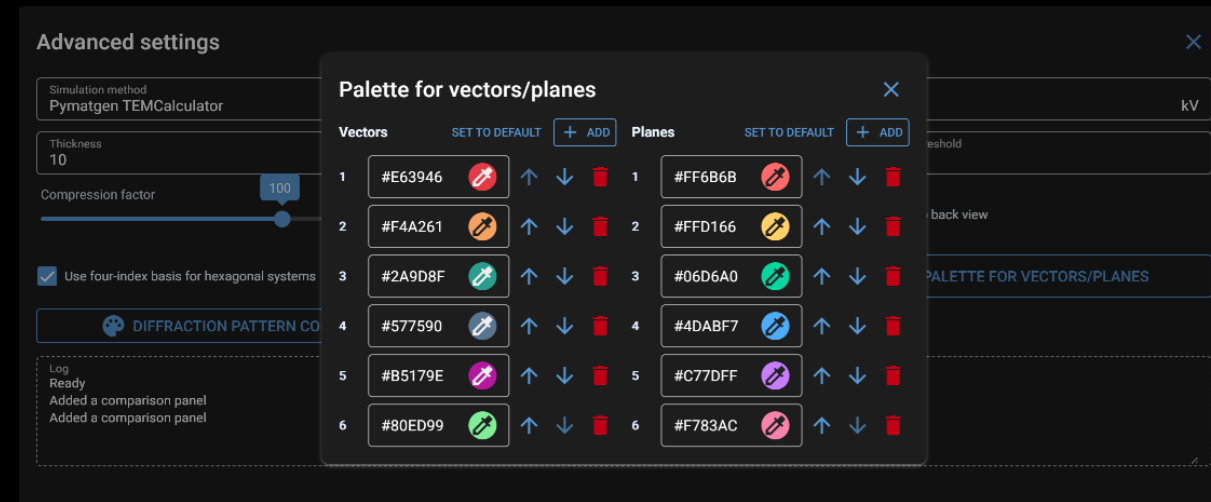
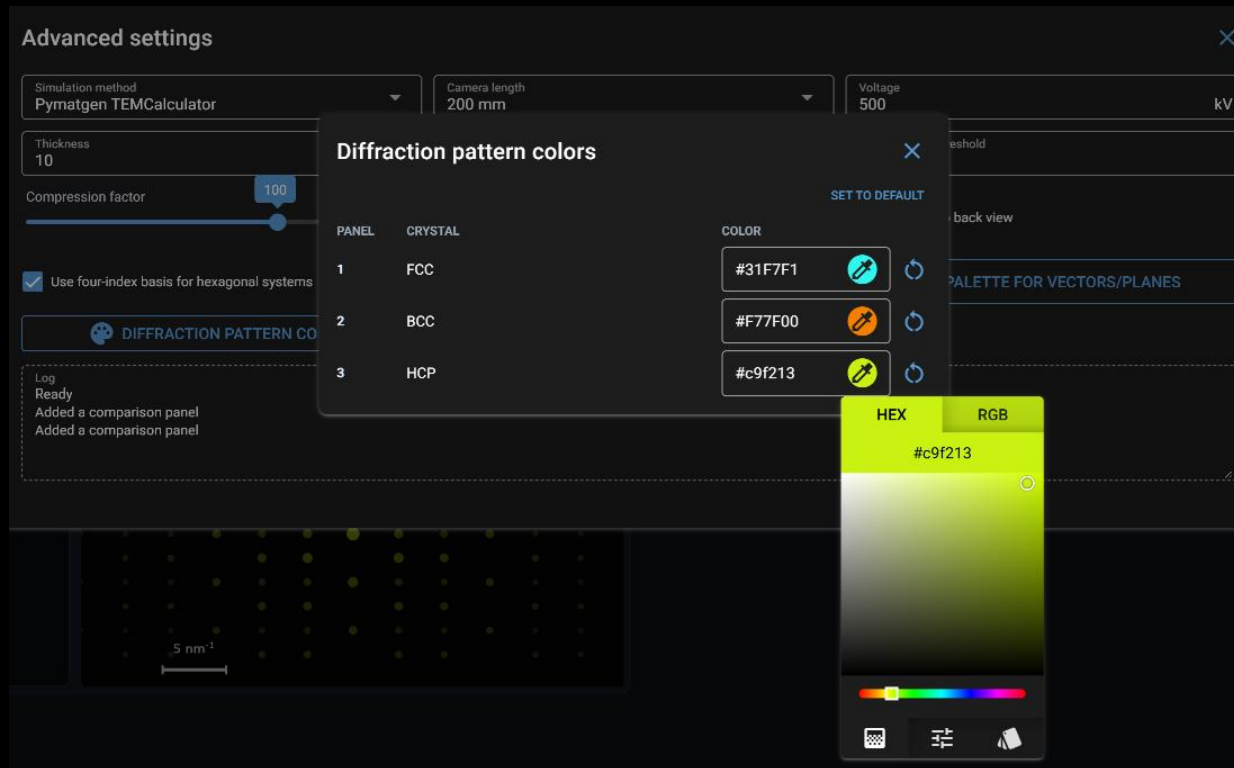
“Always snap back view” is unchecked



The in-plane 45 deg rotation history is retained

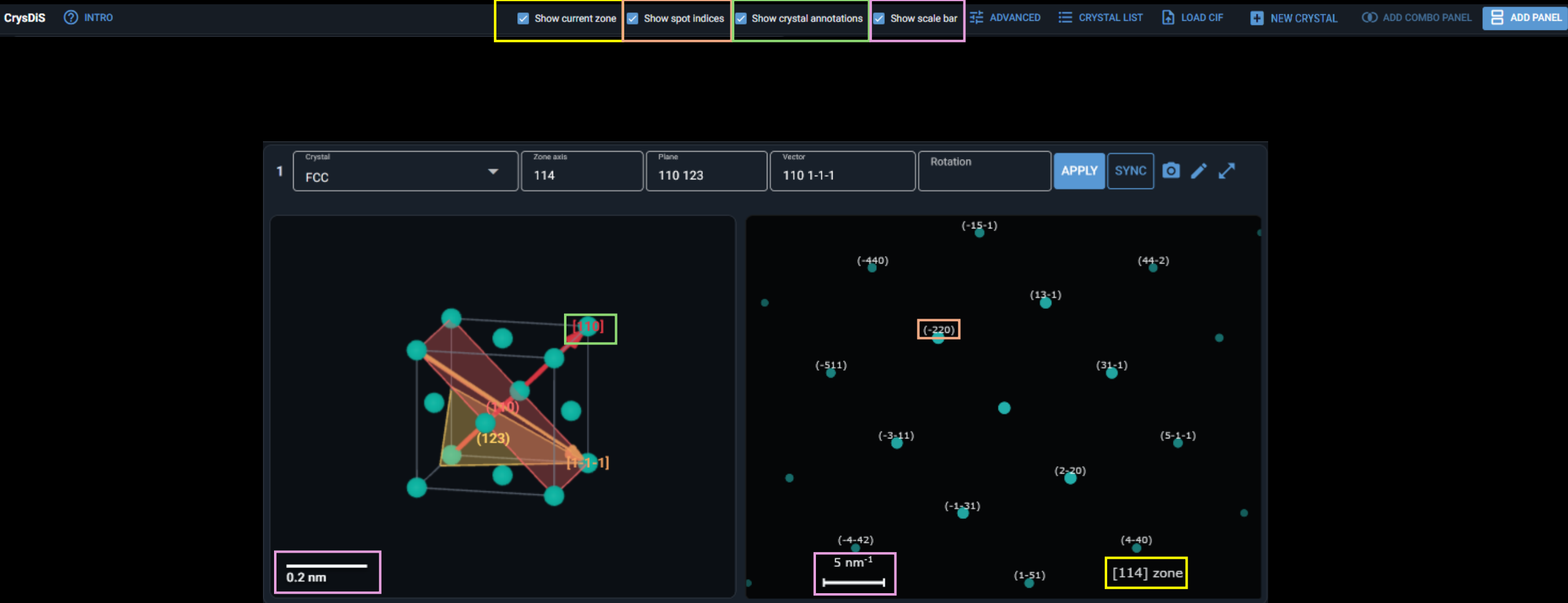
More about advanced settings

- You can also set the colors of diffraction patterns and crystal vectors/planes



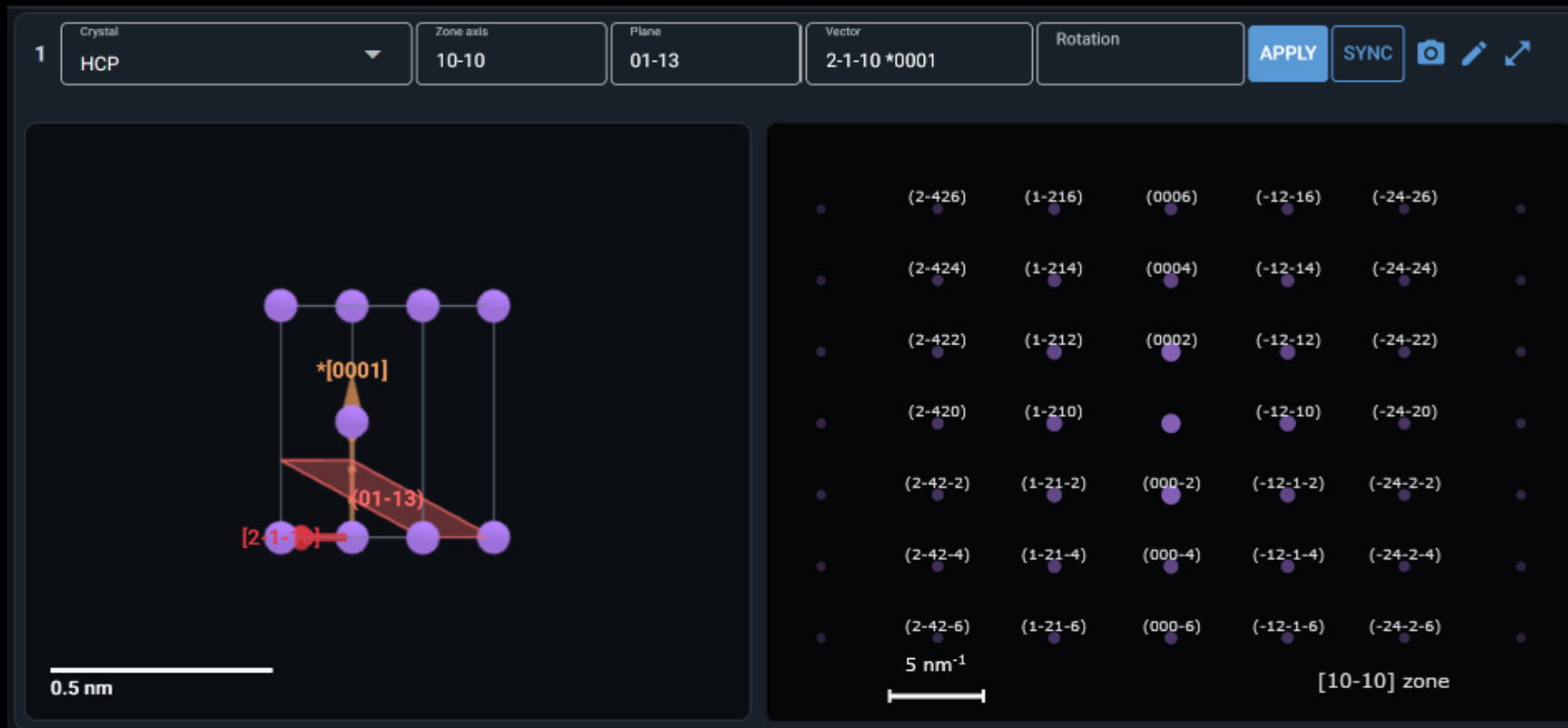
Features of the top toolbar

Some annotation toggles



Features of the panel toolbar

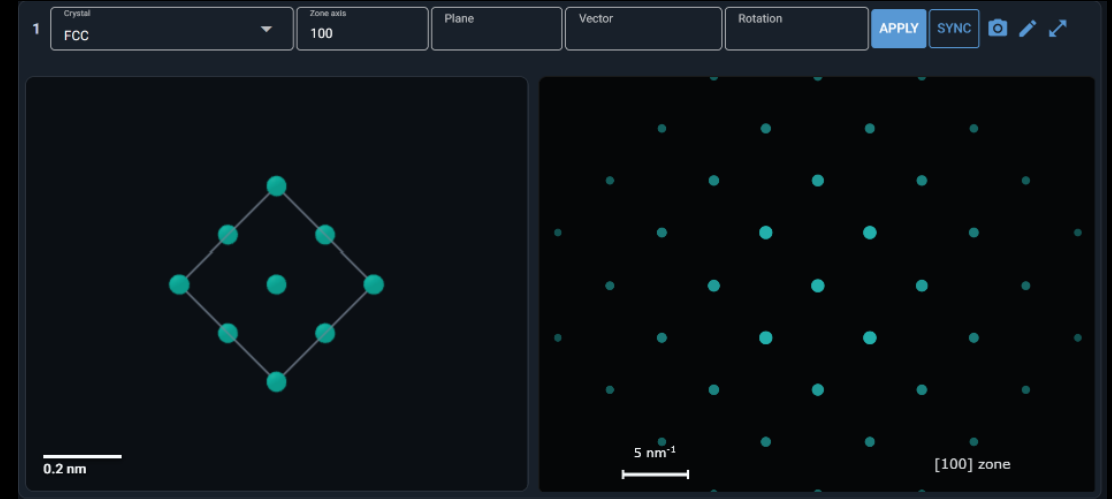
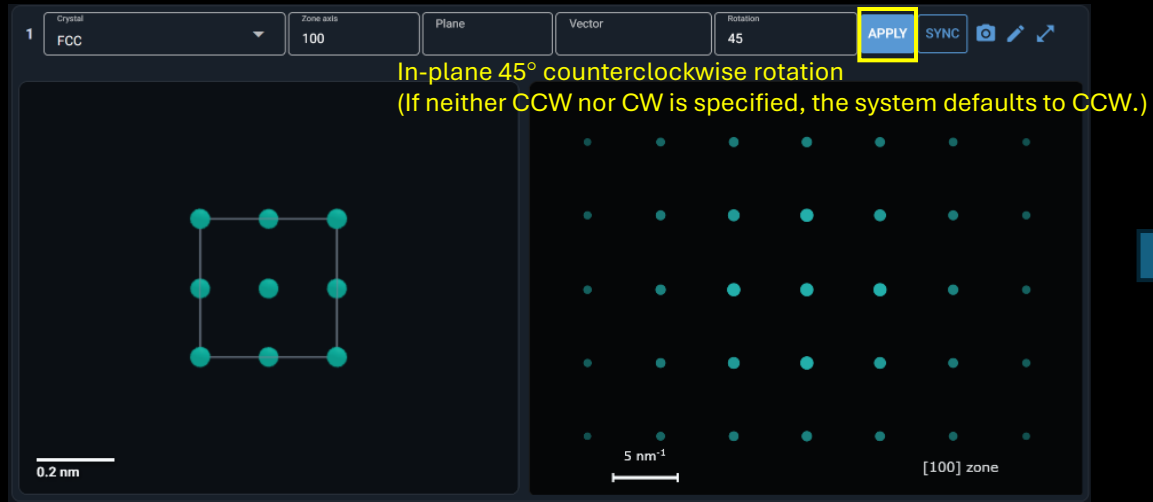
- Choose any crystals in the list or upload/build a new one
- You can go to any zone axis, create vectors/planes
- Reciprocal lattice vectors can be created using *hkl (ex. *110, *0-10)
- For hexagonal systems, 4-index basis inputs are allowed (ex. 2-1-10, *0001)



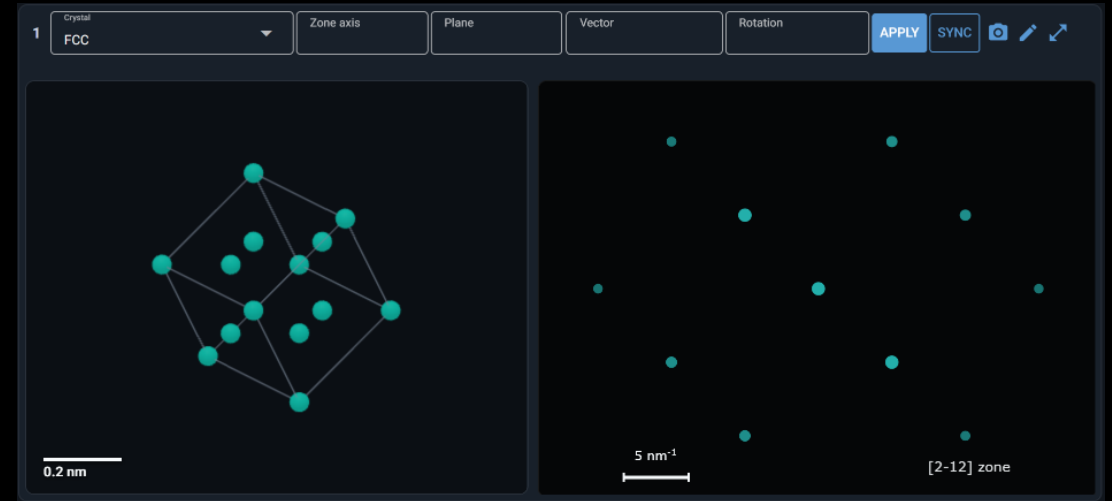
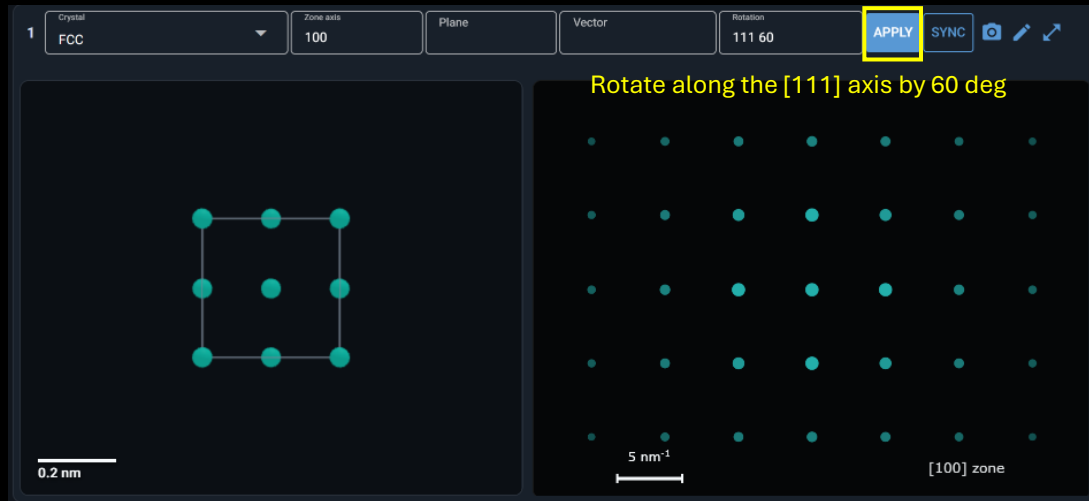
Features of the panel toolbar

- You can also rotate the crystal along a certain axis (right-hand rule is applied)

In-plane rotation: use “deg”, “deg cw”, or “deg ccw”

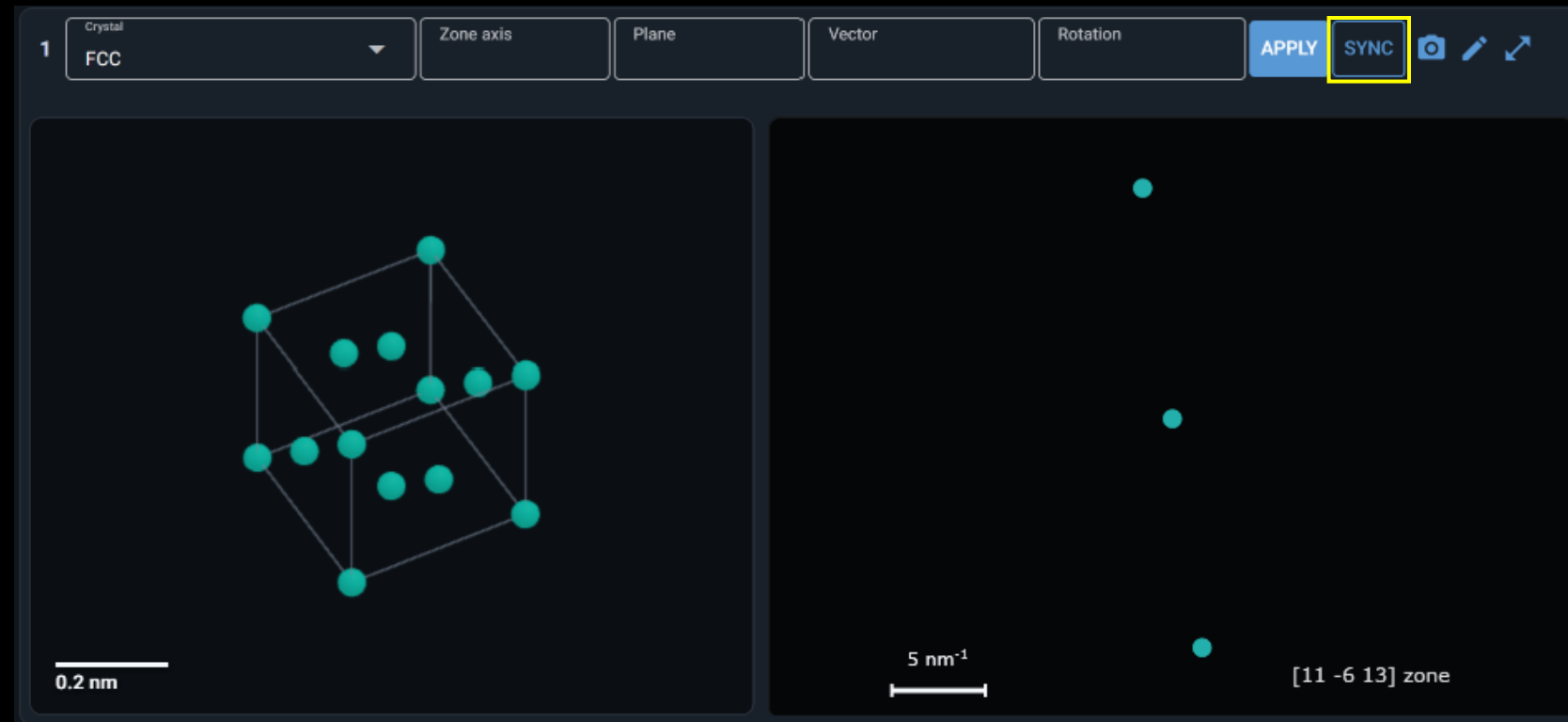


Rotation along a certain axis: use “vector deg”



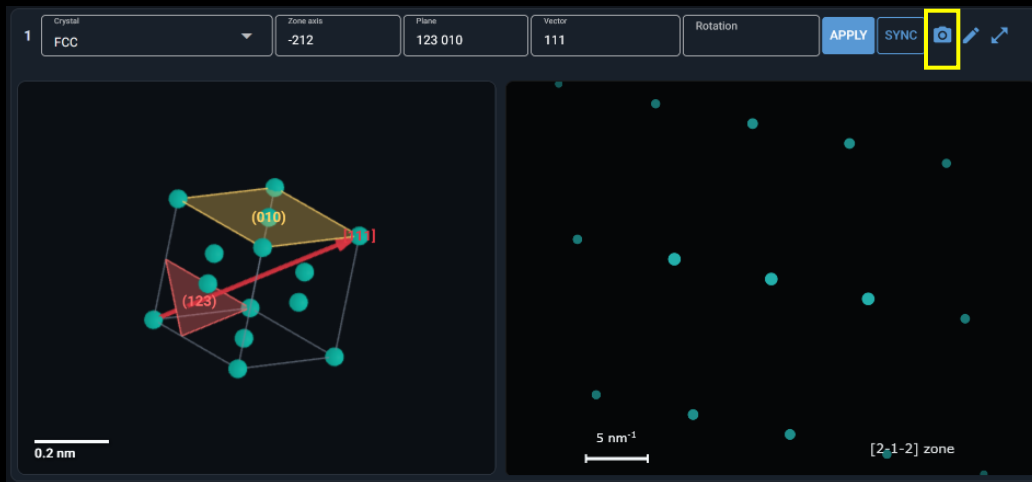
Features of the panel toolbar

- Synchronize the crystal and diffraction pattern using “SYNC”
- It will detect the closest zone axis
- You can also turn on the “Auto sync crystal/diffraction” in the advanced settings



Features of the panel toolbar

- Download the crystal/diffraction pattern at the current viewing angle

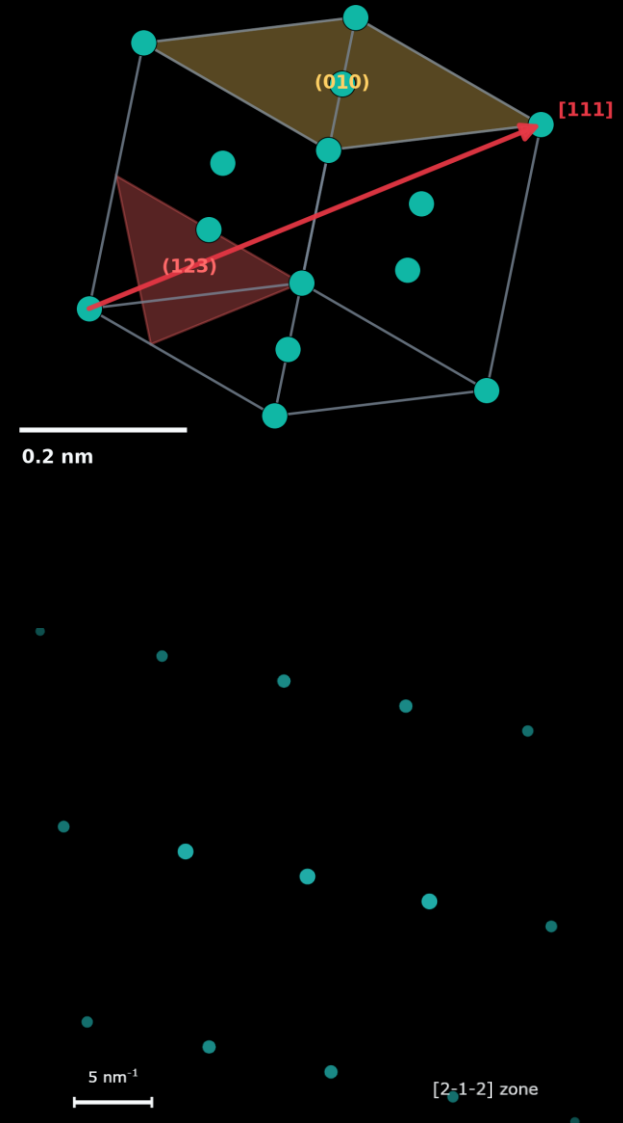


Download panel 1

☒ Real-space crystal
☒ Diffraction pattern
☒ Transparent background

Quality
300 dpi

Export folder: C:\Users\haoen\Downloads



Features of the panel toolbar



- Edit the current structure

Customized crystal builder

Name
L10 MnNi

Lattice
tetragonal

Symmetry
123: P4/mmm

Save target
User app-data library

a
0.374 nm

b
0.374 nm

c
0.352 nm

alpha
90 deg

beta
90 deg

gamma
90 deg

APPLY LATTICE CONSTRAINTS

ADD ATOM SITE

PREVIEW SYMMETRY

APPLY SAME COLOR TO SAME ELEMENT

SET ELEMENT COLOR TO DEFAULT

Atom sites

Atom Mn	x 0	y 0	z 0	Occ. 1	Color #dede00		Site Mn1	
Atom Mn	x 0.5	y 0.5	z 0	Occ. 1	Color #dede00		Site Mn2	
Atom Ni	x 0.5	y 0	z 0.5	Occ. 1	Color #ed02e9		Site Ni1	

SAVE EDITED STRUCTURE

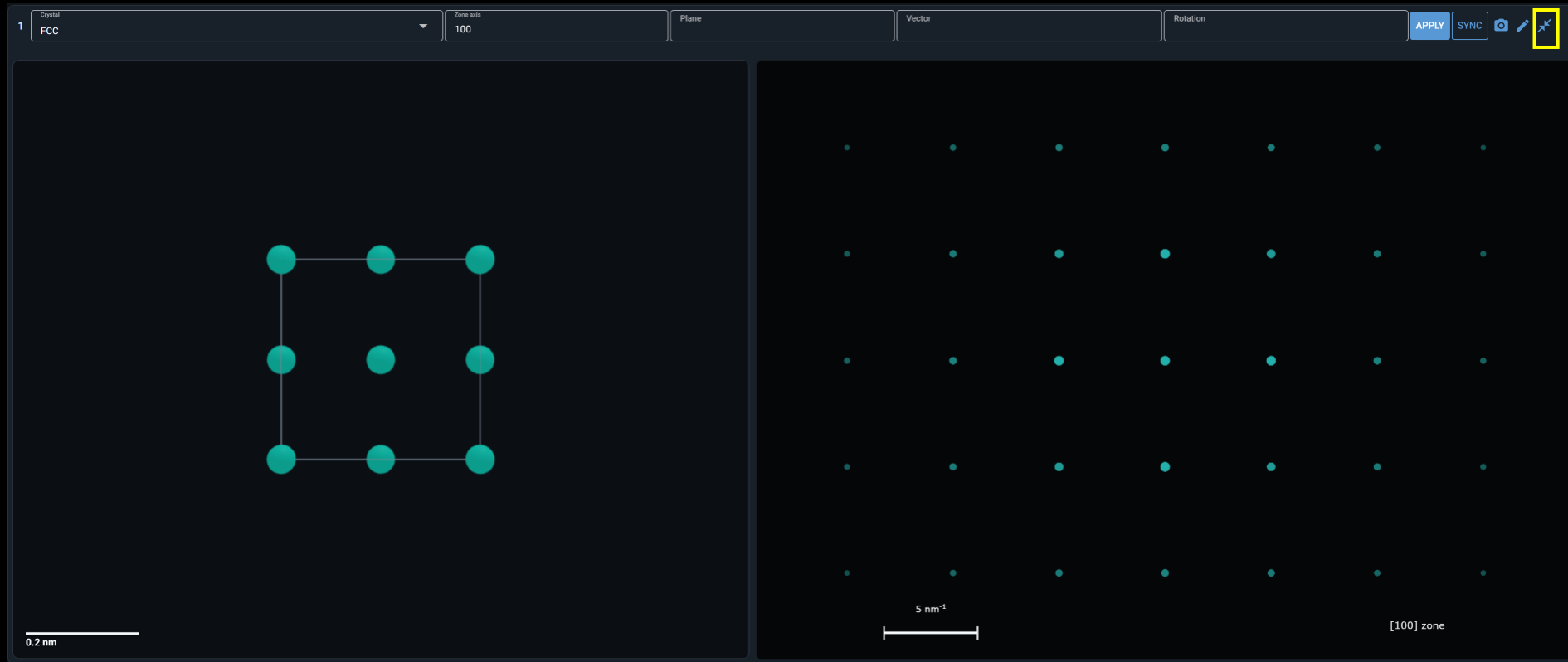
SAVE AS A NEW STRUCTURE

EXPORT AS CIF

Features of the panel toolbar



- Magnify the panel



- Shrink the panel back

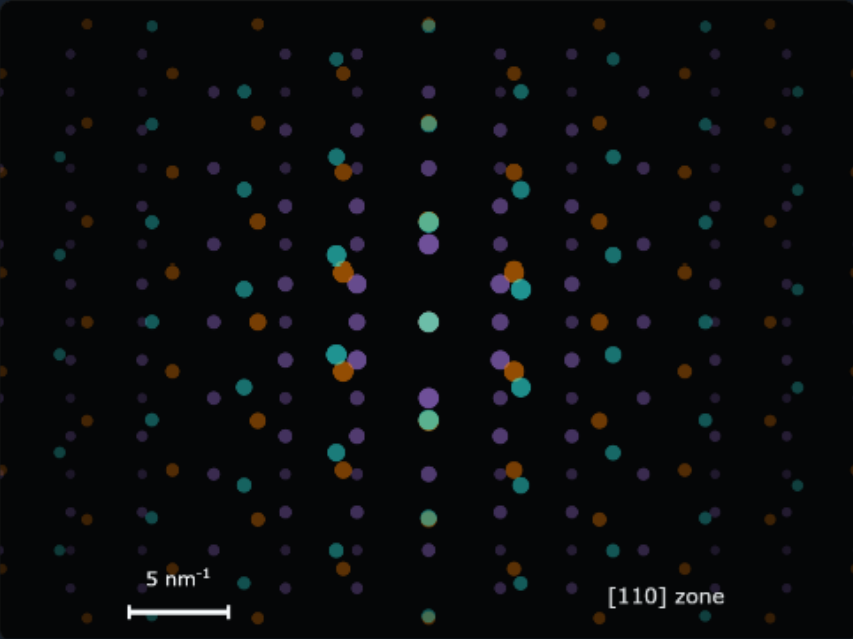
Features of the combo panel

Select crystals present on the page

Add or remove it

C1 Current crystals 1. FCC ADD REMOVE ☐ Bind crystal motion   

Panel	Crystal	Zone	
1	FCC	[110]	↑ ↓ ×
2	BCC	[111]	↑ ↓ ×
3	HCP	[2-1-10]	↑ ↓ ×



5 nm⁻¹ [110] zone

The image shows a software interface for managing crystal diffraction patterns. On the left, a table lists three crystals: FCC (Panel 1, Zone [110]), BCC (Panel 2, Zone [111]), and HCP (Panel 3, Zone [2-1-10]). Each entry has up, down, and delete arrows. Above the table is a dropdown menu showing 'Current crystals' with '1. FCC' selected, and buttons for 'ADD' and 'REMOVE'. To the right of the table is a large rectangular area displaying a 2D diffraction pattern of colored dots (purple, teal, orange) on a dark background. A scale bar at the bottom left indicates 5 nm⁻¹, and a label at the bottom right identifies the pattern as the '[110] zone'. At the top right of the interface are icons for binding crystal motion, a camera, a share icon, and a close button.

Ex. FCC diffraction pattern will be shown on top

The current zone shows the zone axis of the crystal on the top-most layer

The order of the crystal in the list determines the order of the diffraction patterns

Features of the combo panel

Check “Bind crystal motion” if you want the crystals to synchronize their motions

C1

Current crystals
1. FCC

ADDREMOVE☒ Bind crystal motion

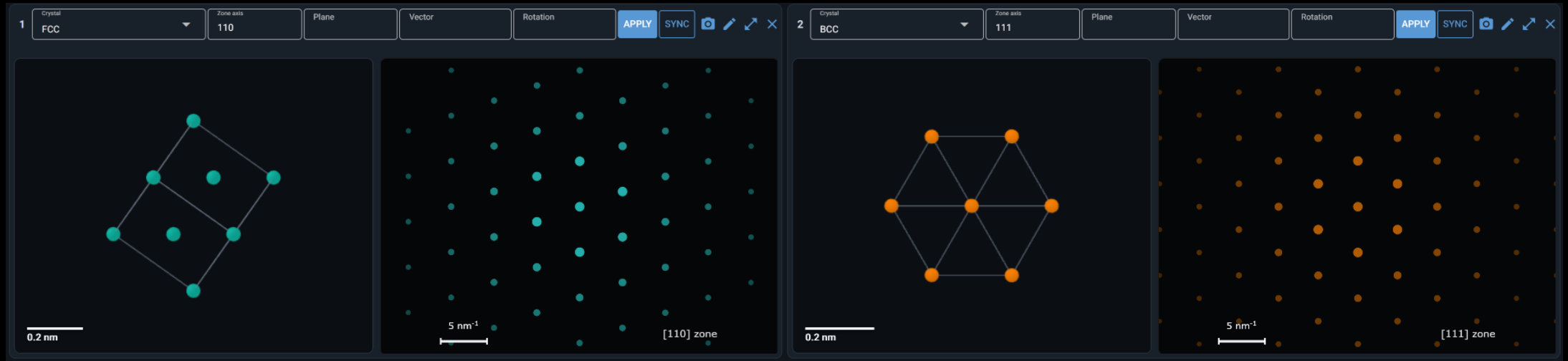
Panel	Crystal	Zone	
1	FCC	[110]	↑ ↓ ×
2	BCC	[111]	↑ ↓ ×

5 nm⁻¹

[110] zone

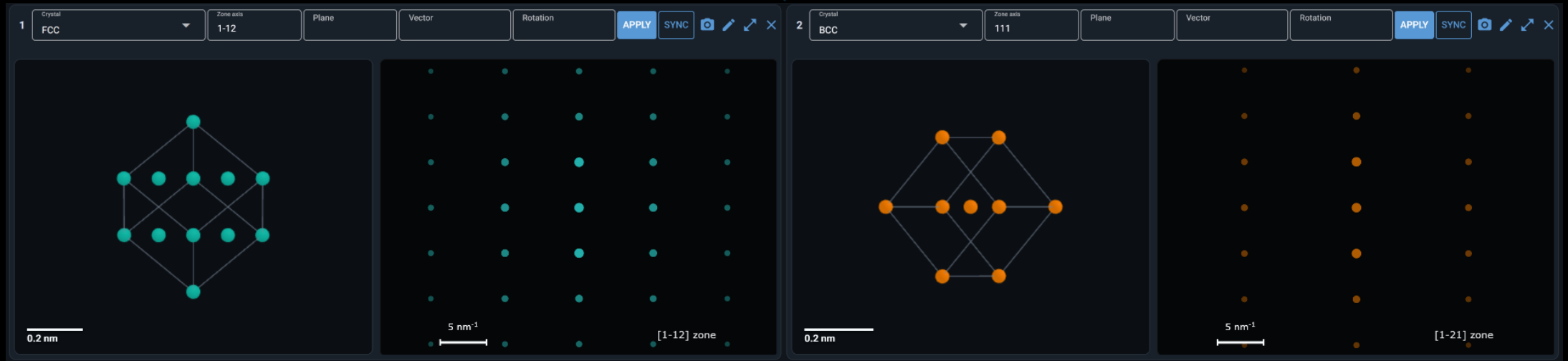
Features of the combo panel

- Bind crystal motion



Ex. if you go to FCC 1-12 zone in panel 1

BCC in panel 2 will do the same motion, and the resulting zone is 1-21



In this case, it means crystals 1 & 2 have the orientation relationship: $[110]_{\text{FCC}} // [111]_{\text{BCC}}$

Comparison of two simulation methods

Simulation method	Ewald Sphere Kinematic	Pymatgen TEMCalculator
Main purpose	More TEM-like kinematic spot visibility	Fast crystallographic zone-axis pattern
Uses voltage?	Yes, through electron wavelength and Ewald sphere radius	Only for wavelength calculation
Uses thickness?	Yes	No
Includes relrod / excitation error?	Yes	No
Includes Ewald sphere curvature?	Yes	No
Good for	Showing why spots appear/disappear with tilt/thickness	Clean ideal zone-axis indexing/intensity